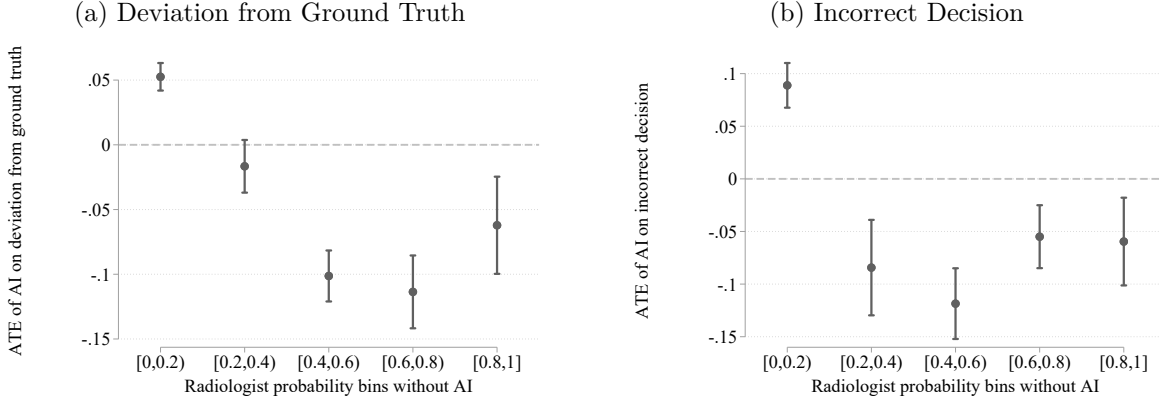


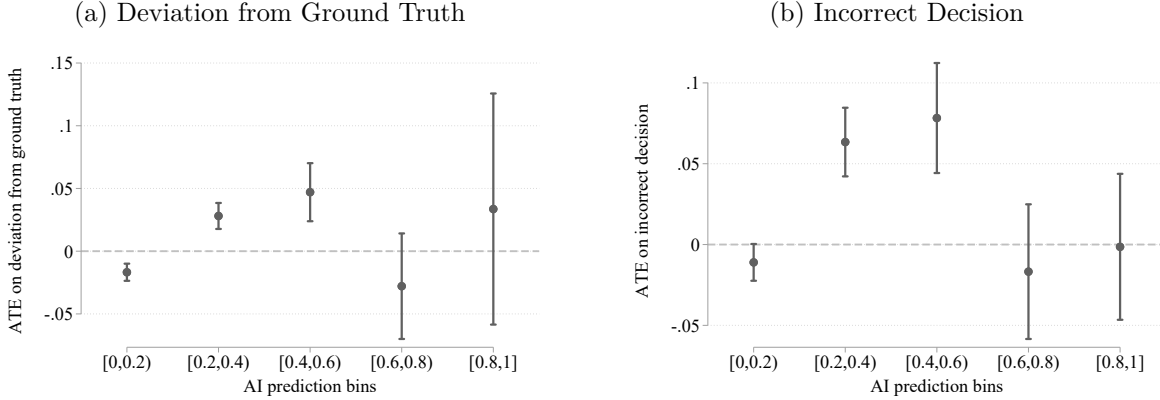
Figure 2: Conditional treatment effect given radiologist prediction



Note: Panel (a) shows the conditional average treatment effect of providing AI information on the absolute value of difference between the radiologist probability and the ground truth. Panel (b) shows analogous treatment effects on incorrect diagnosis, where a correct diagnosis is defined as the treatment recommendation matching the ground truth. Both these treatment effects are conditional on the ranges of the expert’s prediction constructed using the no AI assistance. Standard errors are two-way clustered at the radiologist and patient-case level. The error bars depict 95% confidence intervals. Robustness to experimental design is in appendix C.4.1.

While AI assistance can help uncertain experts, we find that providing uncertain AI predictions reduces performance. As with the analysis of conditional treatment effects given expert predictions, we estimate conditional treatment effects given AI predictions by partitioning cases into five bins based on $\pi(\omega_i = 1 | s_i^A)$. Figure 3 presents the estimates, pooling data from all three experimental designs. When the AI provides a confident prediction (e.g. either close to zero or close to one) performance is significantly improved. We see that in the lowest bins of AI signals, the deviation from the ground truth is reduced. In the second highest bin we also see a marked, though not statistically significant, improvement in performance. However, in the middle range of signals, where the confidence of the AI is low, radiologists’ diagnostic performance and probability of making a correct decision is lower when AI information is provided. This result reinforces the conclusion that radiologists err when updating their beliefs.

Figure 3: Conditional treatment effect given AI prediction



Note: Panel (a) shows the conditional average treatment effect of providing AI information on the absolute value of difference between the radiologist probability and the ground truth. Panel (b) shows analogous treatment effects on incorrect diagnosis, where a correct diagnosis is defined as the treatment recommendation matching the ground truth. Both these treatment effects are conditional on the ranges of AI prediction. Standard errors are two-way clustered at the radiologist and patient-case level. The error bars depict 95% confidence intervals. Robustness to experimental design is in appendix C.4.1 and C.4.2.

4.2.3 Treatment Effects on Time Per Case and Proxies of Effort

Finally, we turn our attention to the effects of AI assistance on time taken and the number of unique interaction (clicks) as a proxy for effort. One hypothesis is that AI assistance could economize on costly human effort without sacrificing overall performance by enabling quicker assessments. At the polar opposite, it is possible that humans take more time because they are provided with more information to process. Which of these effects dominate determines the effect on labor costs when humans use AI assistance, and therefore the optimality of delegating cases versus a collaborative setup.

Our results indicate that radiologists are slower when provided with AI assistance. The last panel of table 2 shows the treatment effects on time spent per case and clicks. These outcomes are measured at the case level. In the X-Ray Only treatment, radiologists spend about 2.6 minutes per case. Both AI and CH increase the time spend per case by a statistically significant amount of approximately 4%. The interaction effect $\gamma_{AI \times CH}$ is not significant for either of the two outcome variables. These effects suggest that decisions where both radiologists and the AI are involved come at a non trivial increase in time spent per case. This result further undercuts the potential benefits in performance from including humans assisted with AI predictions “in the loop.”

4.3 Discussion

We find that AI assistance does not improve the performance of our participants on average, even though the AI predictions are more accurate than the majority of radiologists, and that radiologists respond to this assistance. However, the average treatment effects mask important heterogeneity – AI assistance improves performance for a large set of cases, but also decreases performance in many instances. These results point to biases in the use of AI predictions which we will further investigate in section 5.

We also find that humans have access to valuable contextual information, suggesting that full automation has its drawbacks. But, the biases above – especially in conjunction with our finding that radiologists take longer when given AI assistance – undercut this potential information advantage in a setup that involves AI assistance. Thus, the problem of how best to deploy AI assistance may be non-trivial and the optimal solution may involve selective automation and/or AI assistance. Section 6 analyzes this problem.

Appendix C.4 shows that the results are qualitatively robust to a variety of alternative analyses. The results are similar when we split the analysis by experimental design and when we only focus on the initial across-comparison (using design 1 and 2), although the latter leads to estimates that are imprecise. Alternative ground-truth measures – including a leave-one-out ground truth based on the assessments of our experimental participants and a continuous ground truth, which simply averages the assessments of the ground-truth labelers – also yield similar conclusions. The qualitative patterns of the treatment effects are unchanged if we calibrate each radiologists’ assessments to the ground-truth before conducting the analysis. Finally, incentives for accuracy, which are cross-randomized in designs 1 and 3, do not have significant effects either.

5 Automation/Own-Information Bias/Neglect

An upshot of the results in section 4 is that our participants deviate from the baseline of a Bayesian with correct beliefs about the joint distribution of their own information and the AI signal. In this section, we model and estimate systematic deviations from this benchmark – which we will refer to as Bayesian²⁵ for short – and determine the implications of these deviations on utilizing human expertise and AI predictions.

²⁵The omission of the qualifier “with correct beliefs” slightly abuses terminology because a possible explanation of the deviations we have documented is that our participants are Bayesians but update their beliefs using an incorrect model for the joint distribution of s^A , s^E and ω . We will entertain this possibility below.

5.1 A Model of Deviations from Bayesian Updating

The framework in section 2 shows that a key question is whether the odds-ratios

$$\frac{p_r(\omega_i = 1 | s_i^A, s_{ir}^E)}{p_r(\omega_i = 0 | s_i^A, s_{ir}^E)} \text{ and } \frac{\pi_r(\omega_i = 1 | s_i^A, s_{ir}^E)}{\pi_r(\omega_i = 0 | s_i^A, s_{ir}^E)}$$

differ from each other. Recall that Bayes' rule implies that

$$\ln \frac{\pi_r(\omega_i = 1 | s_i^A, s_{ir}^E)}{\pi_r(\omega_i = 0 | s_i^A, s_{ir}^E)} = \ln \frac{\pi_r(s_i^A | \omega_i = 1, s_{ir}^E)}{\pi_r(s_i^A | \omega_i = 0, s_{ir}^E)} + \ln \frac{\pi_r(\omega_i = 1 | s_{ir}^E)}{\pi_r(\omega_i = 0 | s_{ir}^E)}. \quad (4)$$

We now consider a set of models of belief updating to describe systematic deviations from this benchmark. In our model, the human correctly interprets their own signal when AI assistance is not available but errs when both s_i^A and s_{ir}^E are observed. As we will show below, AI assistance nonetheless unambiguously improves performance for a subset of parameters within this family.

The first class of biases that we consider, arises when the two terms on the right-hand side of equation (4) are incorrectly weighted. Following Grether (1980; 1992), we parametrize this type of error using the following parsimonious functional form:

$$\log \frac{p_r(\omega_i = 1 | s_i^A, s_{ir}^E)}{p_r(\omega_i = 0 | s_i^A, s_{ir}^E)} = b_r \log \frac{\pi_r(s_i^A | \omega_i = 1, s_{ir}^E)}{\pi_r(s_i^A | \omega_i = 0, s_{ir}^E)} + d_r \log \frac{\pi_r(\omega_i = 1 | s_{ir}^E)}{\pi_r(\omega_i = 0 | s_{ir}^E)}, \quad (5)$$

where $b_r, d_r \geq 0$. The Bayesian (with correct beliefs) is a special case with $b_r = d_r = 1$. While this linear form is restrictive, it has been useful for documenting several empirical regularities showing deviations from Bayesian updating like base-rate neglect and under inference (see Benjamin, 2019, for a review).

We will say that the human exhibits *automation bias* if $b_r > d_r$, and *automation neglect* if $b_r < d_r$. As a motivation for this nomenclature, observe that when $b_r > d_r$, the human over-weights the AI signal relative to their own. In the specific case when $d_r = 1$, the agent overshoots when updating the posterior odds relative to a Bayesian. Analogously, if $b_r < d_r$, then the human under-weights the AI signal relative to their own. We say that the human exhibits *own-information neglect* if $d_r < 1$ and *own-information bias* if $d_r > 1$, where the cutoff value of one is based on the Bayesian benchmark. Own-information biases are similar to base-rate biases, but apply to beliefs given the expert's signals instead of unconditional population rates (see Griffin and Tversky, 1992; Kahneman and Tversky, 1973).

A second class of deviations we consider will allow for models in which decision-makers do